

Product category management

Pepperfry Furniture E-Commerce Database Management System

1.Database Design

1.1 Category Table

Attribute	Data Type	Constraint
Category_ID	NUMBER	Primary Key
Category_Name	VARCHAR2(50)	NOT NULL, UNIQUE
Description	VARCHAR2(100)	—

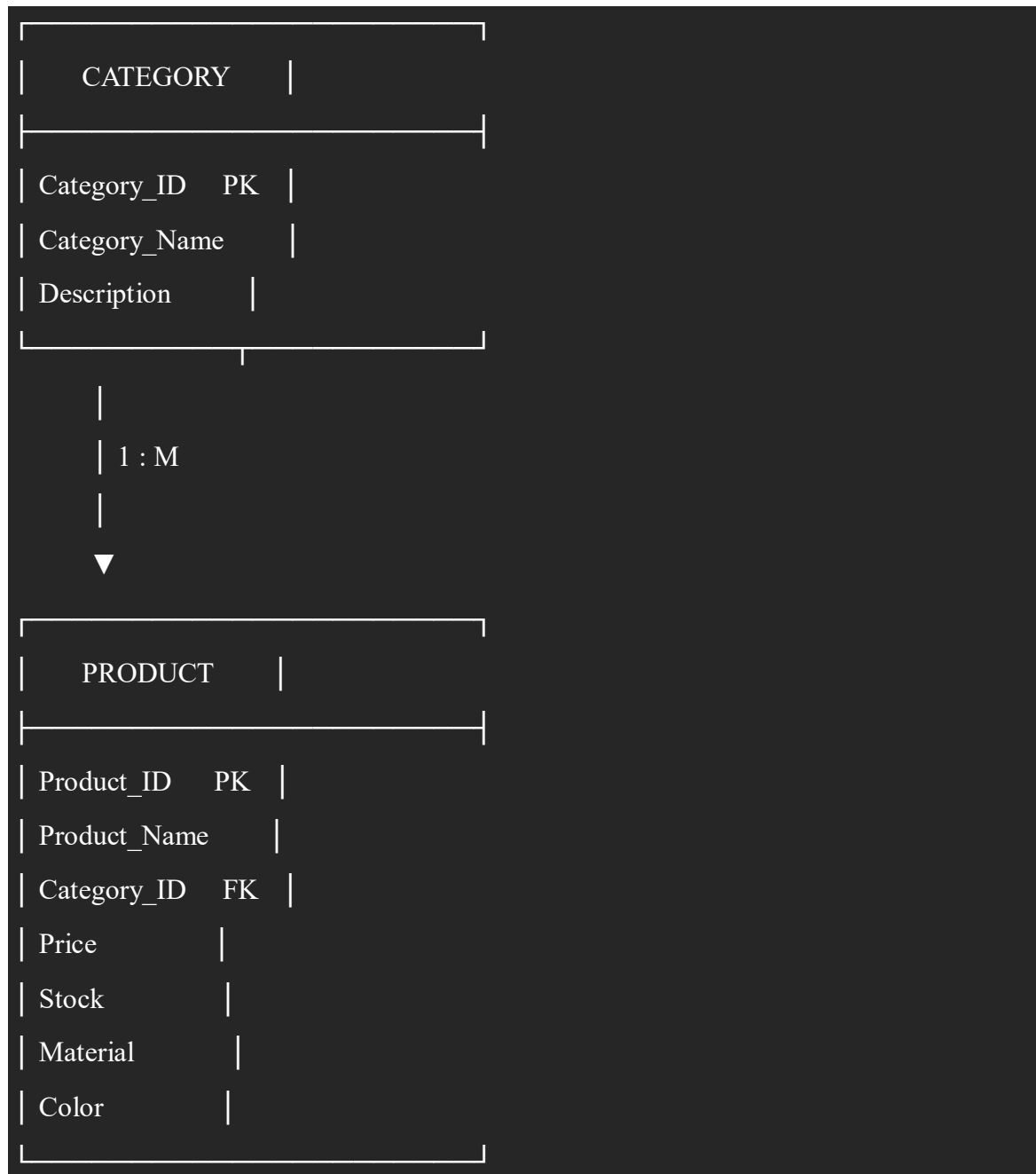
1.2 Product Table

Attribute	Data Type	Constraint
Product_ID	NUMBER	Primary Key
Product_Name	VARCHAR2(100)	NOT NULL
Category_ID	NUMBER	Foreign Key
Price	NUMBER(10,2)	CHECK (> 0)
Stock	NUMBER	CHECK (>= 0)
Material	VARCHAR2(50)	—
Color	VARCHAR2(30)	—

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key
- Product.Category_ID references Category.Category_ID
- Relationship: One-to-Many (1:M)



3 Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID NUMBER PRIMARY KEY,  
3     Category_Name VARCHAR2(50) UNIQUE,  
4     Description VARCHAR2(100)  
5 );
```

Table created.

3.2 Create Product Table

```
SQL> CREATE TABLE Product (  
 2     Product_ID NUMBER PRIMARY KEY,  
 3     Product_Name VARCHAR2(100) NOT NULL,  
 4     Category_ID NUMBER,  
 5     Price NUMBER(10,2) CHECK (Price > 0),  
 6     Stock NUMBER CHECK (Stock >= 0),  
 7     Material VARCHAR2(50),  
 8     Color VARCHAR2(30),  
 9     FOREIGN KEY (Category_ID)  
10     REFERENCES Category(Category_ID)  
11 );
```

Table created.

4. Data Insertion

4.1 Category Records

```
SQL>  
SQL> INSERT INTO Category VALUES  
 2 (1, 'Sofa', 'Sofas and couches');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category VALUES  
 2 (2, 'Bed', 'Bedroom furniture');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category VALUES  
 2 (3, 'Chair', 'Home and office chairs');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category VALUES  
 2 (4, 'Table', 'Dining and study tables');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category VALUES  
 2 (5, 'Storage', 'Storage furniture');  
  
1 row created.
```

4.2 Product Records

```

SQL> INSERT INTO Product VALUES
  2 (101, 'Luxury Sofa', 1, 35000, 10, 'Wood', 'Brown');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (102, 'Modern Sofa', 1, 28000, 15, 'Fabric', 'Grey');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (103, 'King Size Bed', 2, 45000, 8, 'Wood', 'Walnut');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (104, 'Queen Size Bed', 2, 38000, 12, 'Wood', 'White');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (105, 'Office Chair', 3, 8500, 25, 'Plastic', 'Black');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (106, 'Dining Chair', 3, 5000, 30, 'Wood', 'Brown');

```

5. Data Retrieval

5.1 Display Category Records

```
SQL> SELECT * FROM Category;
```

CATEGORY_ID	CATEGORY_NAME
-------------	---------------

DESCRIPTION

1 Sofa
Sofas and couches

2 Bed
Bedroom furniture

3 Chair
Home and office chairs

CATEGORY_ID	CATEGORY_NAME
-------------	---------------

DESCRIPTION

4 Table
Dining and study tables

5 Storage
Storage furniture

5.2 Display Product Records

```
SQL> SELECT * FROM Product;
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
CATEGORY_ID    PRICE    STOCK
```

```
MATERIAL
```

```
COLOR
```

```
101  
Luxury Sofa  
1 35000 10
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
CATEGORY_ID    PRICE    STOCK
```

```
MATERIAL
```

```
COLOR
```

```
Wood  
Brown
```

```
PRODUCT_ID
```

6. Product Update

Updated Product Records – Price, Stock and Color Changes

```
SQL> UPDATE Product  
2 SET Price = 37000  
3 WHERE Product_ID = 101;
```

```
1 row updated.
```

```
SQL> UPDATE Product  
2 SET Stock = 20  
3 WHERE Product_ID = 105;
```

```
1 row updated.
```

```
SQL> UPDATE Product  
2 SET Color = 'Blue'  
3 WHERE Product_ID = 108;
```

```
1 row updated.
```

7. Product Deletion

Deleted Product Records – Wooden Cupboard and Storage Cabinet

```
SQL>
SQL> DELETE FROM Product
  2  WHERE Product_ID = 109;

1 row deleted.

SQL> DELETE FROM Product
  2  WHERE Product_ID = 110;

1 row deleted.
```

8. Category-wise Product Reports

```
SQL> SELECT
  2     Category.Category_Name,
  3     Product.Product_Name,
  4     Product.Price,
  5     Product.Stock
  6 FROM Category
  7 JOIN Product
  8 ON Category.Category_ID = Product.Category_ID
  9 ORDER BY Category.Category_Name;
```

CATEGORY_NAME

PRODUCT_NAME

	PRICE	STOCK
Bed		
Queen Size Bed	38000	12

Bed		
King Size Bed	45000	8

CATEGORY_NAME

PRODUCT_NAME

	PRICE	STOCK
Chair		
Office Chair		

Category-wise Total Product Count

```
SQL> SELECT
  2     Category.Category_Name,
  3     COUNT(Product.Product_ID) AS Total_Products
  4 FROM Category
  5 LEFT JOIN Product
  6 ON Category.Category_ID = Product.Category_ID
  7 GROUP BY Category.Category_Name
  8 ORDER BY Category.Category_Name;
```

CATEGORY_NAME	TOTAL_PRODUCTS
Bed	2
Chair	2
Sofa	2
Storage	0
Table	2